

AMAR DHILLON

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Education

University of Maryland, College Park

B.S. in Computer Science, Minor in Statistics; GPA: 3.5

Expected May 2028

College Park, MD

Experience

Amazon Web Services

Sep 2026 – Dec 2026

Software Development Engineer Intern

Redmond, WA

- Writing Java logic that sets the upper and lower bounds on EC2 capacity Amazon Redshift reserves for operations like patching, balancing availability against cost, and backtesting it

Capital One

Jun 2026 – Aug 2026

Software Engineering Intern

Richmond, VA

- Cut LLM workflow generation time **65%** to under 15s by splitting one large Claude skill into smaller sub-skills that call Python tools for search and validation, and output quality held in repeated stress-test runs
- Because the 270K+ row PostgreSQL workflow table had strict access limits, precomputed popular workflows into a category tree that the Claude skill searches first, with Redis caching full workflows to cut database queries
- Rebuilt the legacy tool developers use to create test-data workflows as a React and TypeScript Electron app with a visual React Flow editor and Jest-tested validation, and **90%** of 30+ surveyed developers preferred it

UMD App Development Contracting (Client: Amazon Leo)

Jan 2026 – Present

Software Engineer (Contract)

College Park, MD

- Automated retraining for a production Amazon Bedrock classifier that rates research documents by importance: on an EventBridge schedule a larger LLM grades its ratings, and fine-tuning runs only when the grades shift past a threshold
- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda
- Deployed a space-topic research tool for Amazon Leo's public-policy team and set it up for handoff with AWS CDK infrastructure-as-code and GitHub Actions CI/CD, so Leo engineers new to the project now maintain it on AWS Fargate

UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

Machine Learning Research Intern

College Park, MD

- For a PhD-led robotics project aimed at ICLR, built a pipeline that shows a vision-language model AnyGrasp's candidate grasps as Open3D overlays on multi-camera images, then tuned prompts so the model picks the right grasp more often

TerpLabs

Jun 2025 – Present

Co-Founder, Software Engineer

College Park, MD

- Building a RAG meal-suggestion feature for the official UMD app, in which GPT-4o mini writes suggestions from scraped recipes found by SentenceBERT vector search in ChromaDB, and internal testers rated **93%** of meals satisfactory
- Co-founded a 110+ member student product studio that has shipped 5 live products
- Built Redis caching in front of the Prisma ORM queries for Tortuga, the studio's class scheduler with **20,000+ visits** in its first 30 days, to reduce database load and latency

Projects

IEX Order Book & Matching Engine | *C++17, STL, POSIX, Linux perf*

Jul 2026 – Aug 2026

- Built a C++17 order book and matching engine and replayed **50.9M** real IEX exchange messages through it without ever resyncing to the exchange's book, agreeing with the exchange on 99,122 of 99,123 fills and 100% of top-of-book prices

Technical Skills

AI/ML: LLMs (Claude, Amazon Bedrock, GPT-4o mini), retrieval-augmented generation (RAG), vector search (ChromaDB, SentenceBERT), LLM agents and tool use (Claude skills), model fine-tuning and evaluation, vision-language models, prompt engineering, PyTorch, Open3D

Languages: Python, Java, C++, C, TypeScript, JavaScript, SQL, CUDA

Cloud & Data: AWS (Lambda, Kinesis, EventBridge, Fargate, CloudWatch), AWS CDK, monitoring (CloudWatch, X-Ray), REST APIs, PostgreSQL, Redis, MongoDB (NoSQL), OpenSearch

Engineering: full-stack apps (React, Electron), data structures and algorithms, object-oriented design, Docker, GitHub Actions (CI/CD), Linux, unit testing (Pytest, Jest), Git, uv

Certification: AWS Certified Developer – Associate (Nov 2024 – Nov 2027)